Lesson Planning

Complete Lesson Plan Book

Class: Class 8

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Square Roots and Cube Roots

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Unit 1: Square Roots and Cube Roots

Learning Objectives

• Students will define square roots and cube roots and identify perfect squares and perfect cubes.

• Students will find square roots and cube roots of numbers up to 1000 using prime factorisation and estimation.

• Students will solve simple word problems involving square roots and cube roots in everyday situations.

Key Concepts

• The square root of a number is a value that, multiplied by itself, gives the original number; for example, the square root of 49 is 7 because 7 x 7 = 49.

• A perfect square is a number whose square root is a whole number, such as 1, 4, 9, 16, 25, 36, 49, 64, 81 and 100.

• The cube root of a number is a value that, multiplied by itself three times, gives the original number; for example, the cube root of 27 is 3 because 3 x 3 x 3 = 27.

• A perfect cube is a number whose cube root is a whole number, such as 1, 8, 27, 64, 125 and 216.

• Prime factorisation helps find roots: group the prime factors in pairs for a square root and in triples for a cube root.

Teaching Methodology (Step by Step)

1. The teacher writes 5 x 5 = 25 on the board and asks the class what number multiplied by itself gives 25, introducing the square root symbol.

2. The teacher explains perfect squares with a table of squares from 1 to 15 and asks students to read the square roots from it.

3. The teacher shows how to find the square root of 144 by prime factorisation, grouping factors in pairs.

4. The teacher introduces cube roots with a small cube of 27 blocks and shows that 3 x 3 x 3 = 27.

5. The teacher works out the cube root of 216 by grouping prime factors in triples and checks the answer by cubing it.

6. The teacher gives five mixed numbers and asks students to decide whether each has an exact square root, an exact cube root, or neither.

Classroom Activities

• Root race: the teacher calls out numbers and students write the square root on mini whiteboards; the fastest correct answer wins the round.

• Factor hunt: in pairs, students factorise 225 and 343 and use the groups to find their square and cube roots.

• Number sort: students sort a set of number cards into perfect squares, perfect cubes, both, or neither.

Assessment Questions

1. Find the square root of 121 and the cube root of 125.

2. Is 200 a perfect square? Is 200 a perfect cube? Give a reason for each answer.

3. A square garden has an area of 196 square metres. What is the length of one side?

Homework

• Find the square roots of 81, 169 and 256 and the cube roots of 64, 343 and 512 using prime factorisation.

• Write three real-life examples where a square root or cube root is used, such as finding the side of a square plot from its area.

Time Allocation

40 minutes

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